Memory Interface

Every µP-based system has a memory system. All systems contain two types of memories.

- **Read-Only Memory** (Volatile Memory)
- **Random Access Memory** (Read/Write Memory or Non-volatile Memory)

ROM contains system software and permanent system data.

RAM contains system temporary data and system application software.

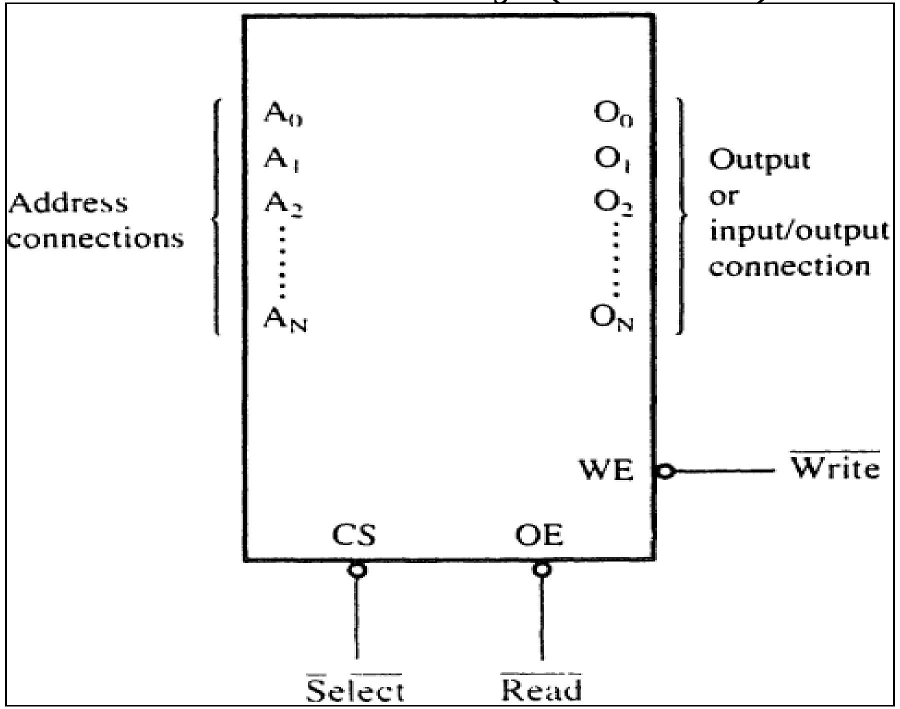
Memory Devices:

- Read-Only Memory
- Flash Memory (EEPROM)
- Static Random Access Memory (SRAM)
- Dynamic Random Access Memory (DRAM)

Memory Pin Connections:

Pin connections common to all memory devices are:

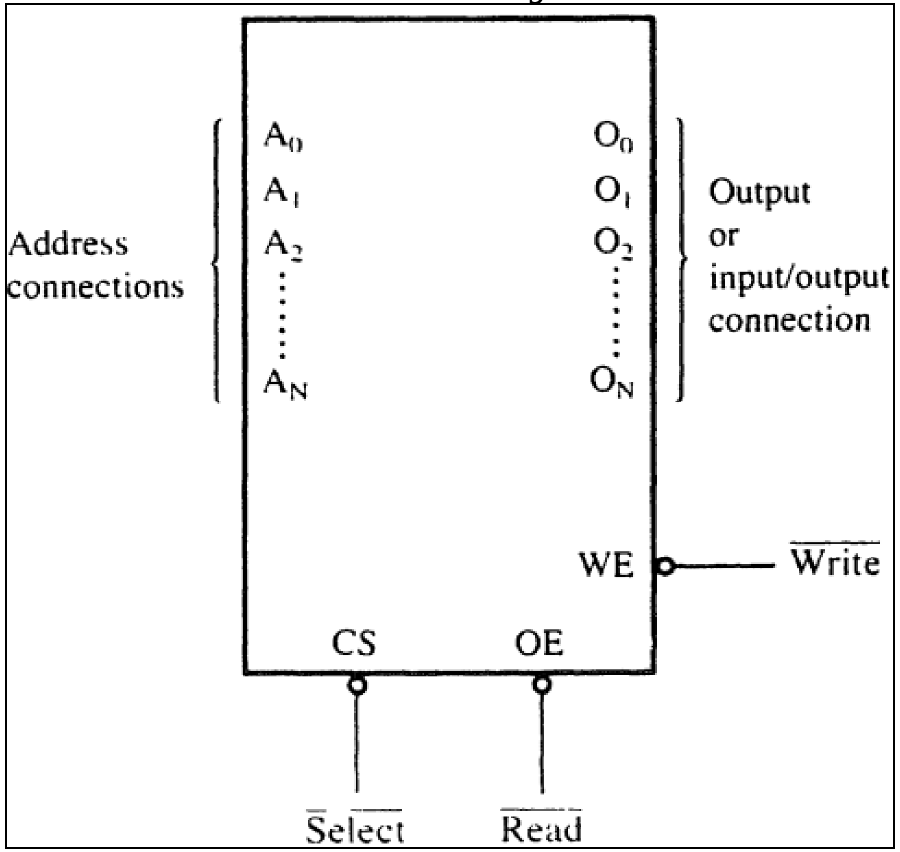
- ✓ Address Inputs
- ✓ Data Outputs or Input(s)/Output(s)
- ✓ Selection Input
- ✓ Control input to selects READ or Write



Address Connections: Address inputs selects a memory location within the memory device.

- ✓ Labeled from $\mathbf{A_0}$, least significant address input
- ✓ A_N, '**N**' labeled, '**N-1**'then the total number of address pins

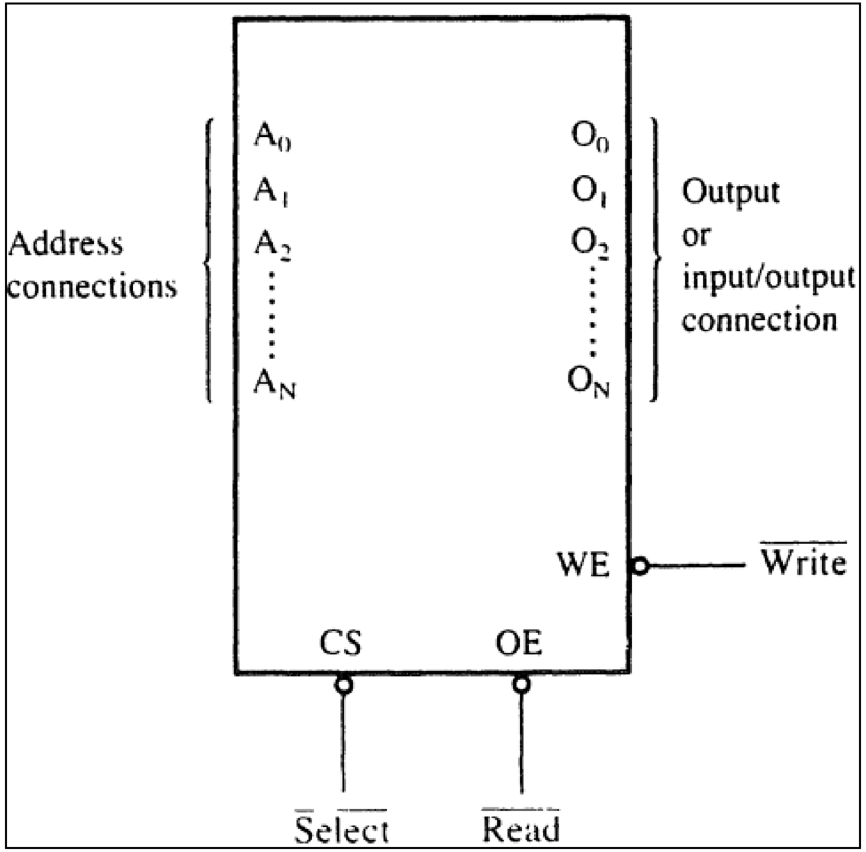
Memory	Address Pins (N)	Address Connections	Memory Locations (2^N)
1K	10	A_9-A_0	1024
2K	11	A_{10}-A_0	2048
4K	12	A_{11}-A_0	4096
8K	13	A_{12}-A_0	8192
1M	20	A_{19}-A_0	1,048,576



Memory Device	Memory Sys. Section	Decoded Starting Address	Last Location Address
1K	400H (1024)	10000H	103FFH (10000H+400H-1H)
4K	1000H (4096)	14000H	14FFFH (14000H+1000H-1H)
64K	10,000H	30000H	3FFFFH
1M	100,000H	-	-

Data Connections:

All memory deices has a set of data outputs or input(s)/output(s). Today, many devices have bi-directional common IO pins.



- ✓ Data pins are labeled with **D_7-D_0** for an **8-bit-wide** memory (Means memory device stores 8-bit of data in each of its Memory locations).
- ✓ 8-bit-wide memory device is often called **byte-wide** memory
- ✓ Mostly devices are **8-bit-wide**, some devices are **16-bits**, **4-bits** or just **1-bit-wide**

- ❖ Datasheet (Catalog Listing) of memory devices often Represented by: **Memory Locations x Bits per Location**

Ex: Memory Device with **1-K** memory locations and **8-bits** in each location is often listed as **1K x 8** by the manufacturer. (**64K x 4 , 16K x 1)**

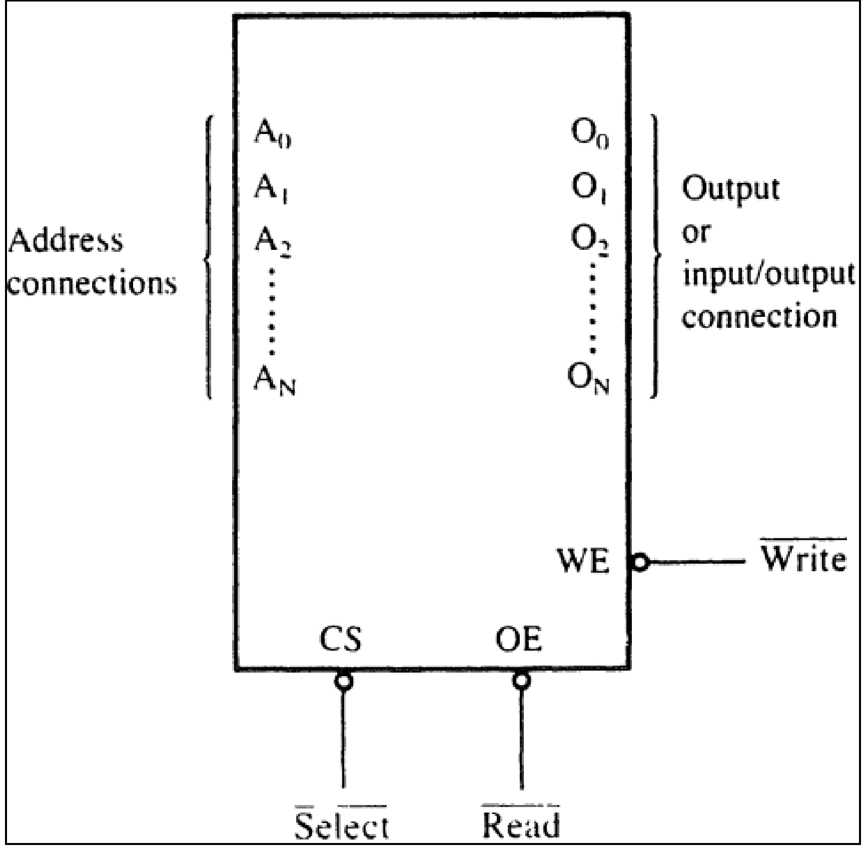
- ❖ Sometime memory devices classified as **Total bit Capacity**.

Ex: **8K, 256K**

Selection Connections:

Memory devices has one or more selection or enable inputs

- Chip Select ($\overline{\mathbf{CS}}$)
- Chip Enable ($\overline{\mathbf{CE}}$)
- Select ($\overline{\mathbf{S}}$)



❖ **RAM** has at least one $\overline{\mathbf{CS}}$ or $\overline{\mathbf{S}}$.

❖ **ROM** has at least one $\overline{\mathbf{CE}}$.

✓ If $\overline{\mathbf{CS}}$**,** $\overline{\mathbf{CE}}$ **or** $\overline{\mathbf{S}}$ input is **active** (**Logic 0**), memory device performs a **read or write** operation

✓ If $\overline{\mathbf{CS}}$**,** $\overline{\mathbf{CE}}$ **or** $\overline{\mathbf{S}}$ input is in**active** (**Logic 1**), memory device **do not** performs a **read or write** operation b/c it is **turned OFF** or **Disabled**.

Control Connections:

Memory devices have control input (s).

- ROM has only one control input
- RAM has one or two control inputs

❖ Control input on a ROM is Output Enable ($\overline{OE}$) or Gate ($\overline{G}$).

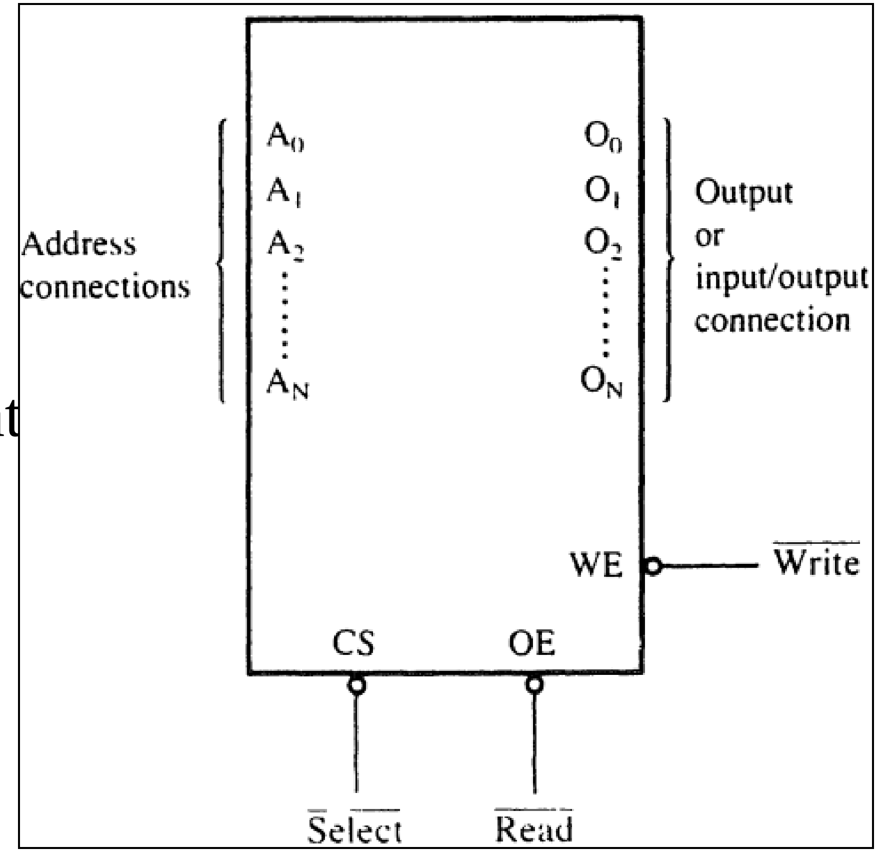
- ✓ If $\overline{OE}$ = 0 and $\overline{CE}$ = 0, output is enabled
- ✓ If $\overline{OE}$ = 1 and $\overline{CE}$ = 0, output is at high-impedance state (disabled)

❖ Control input on a RAM, if one

- ✓ R/$\overline{W}$ and $\overline{CS}$ = 0

Two inputs

- ✓ $\overline{WE}$ (or $\overline{W}$) = 0, to perform memory write
- ✓ $\overline{OE}$ (or $\overline{G}$) = 0, to perform memory read
- ✓ these two pins must not be activated (**Logic 0**) at



ROM Memory:

- ROM most commonly used type is **EPROM** (Erasable Programmable Read-Only Memory)
- **EEPROM** (Electrically Erasable Programmable Read-Only Memory) or **Flash Memory**

Types of EPROM with following part numbers:

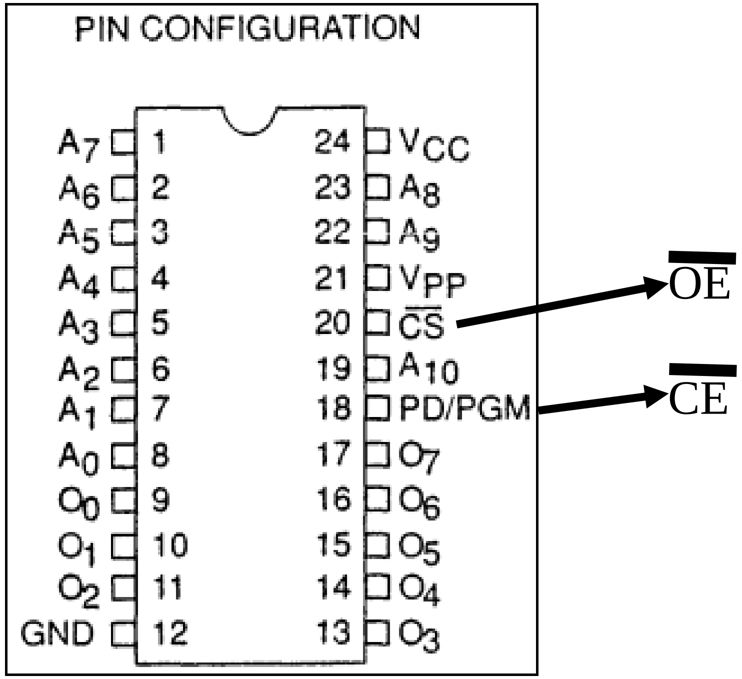
2704	512 x 8
2708	1K x 8
2716	2K x 8
2732	4K x 8
2764	8K x 8
27128	16K x 8
27256	32K x 8
27512	64K x 8
271024	128K x 8

Each has

- ✓ Address inputs
- ✓ Eight data pins
- ✓ One or more Chip Selection inputs ($\overline{CE}$), and output enable ($\overline{OE}$)

Commonly used **EPROM is 2716 EPROM**

- ✓ has 11 address inputs and 8 data outputs
- ✓ It is 2K x 8 memory device



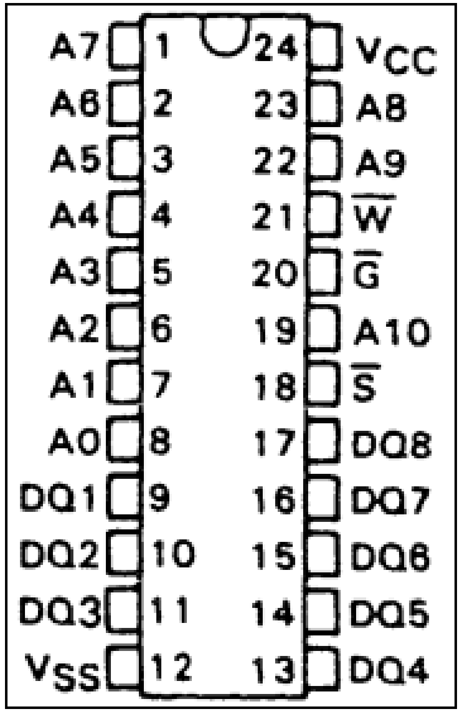
PIN NAMES

A_0–A_{10}	ADDRESSES
PD/PGM	POWER DOWN/PROGRAM
$\overline{CS}$	CHIP SELECT
O_0–O_7	OUTPUTS

PINS / MODE	PD/PGM (18)	$\overline{CS}$ (20)	V_{PP} (21)	V_{CC} (24)	OUTPUTS (9-11, 13-17)
Read	V_{IL}	V_{IL}	+5	+5	D_{OUT}
Deselect	Don't care	V_{IH}	+5	+5	High Z
Power Down	V_{IH}	Don't care	+5	+5	High Z
Program	Pulsed V_{IL} to V_{IH}	V_{IH}	+25	+5	D_{IN}
Program Verify	V_{IL}	V_{IL}	+25	+5	D_{OUT}
Program Inhibit	V_{IL}	V_{IH}	+25	+5	High Z

Static RAM (SRAM):

- ✓ TMS4016 is a 2K x 8 read/write memory
- ✓ 11 address inputs
- ✓ 8 data input/output



PIN NOMENCLATURE	
A0 – A10	Addresses
DQ1 – DQ8	Data In/Data Out
$\overline{G}$	Output Enable
$\overline{S}$	Chip Select
V_{CC}	+5-V Supply
V_{SS}	Ground
$\overline{W}$	Write Enable

Address Decoding:

✓ Its is necessary to decode the address sent from the µP

✓ Without address decoder only one memory can be connected

Why Decode Memory:

8088 is compared to the 2716 EPROM

- 20-bit 11-bit
- 1M x 8 2K x 8

The decoder corrects the mismatch by decoding the address pins that do not connect to the memory device.

For Decoding:

- **NAND Decoder**
- **3-to-8 Line Decoder**

Examples:

- **Design an interface for 8088µp to connect a single 2716 (2Kx8) EEPROM such that memory start at address 00000H.**

Note: Remember these steps for decoding

1. **Write pin numbers of the Microprocessor**
2. **Under the pin numbers of the Microprocessor, write the pin numbers of the Memory and mark them with 'X' under the memory pin number**
3. **Start assigning address from MSB till it reaches the mark 'X'**
4. **Find the Starting address of memory by changing mark 'X' with '0s', and write it in hexadecimal format**
5. **Find the Ending address of memory by changing mark 'X' with '0s', and write it in hexadecimal format**
6. **Draw the decoding circuitry for interfacing microprocessor with a memory 2716 (2Kx8)**

If the 20-bit binary address is decoded by the NAND gate

- ✓ Leftmost 9 bits are 1's
- ✓ Rightmost 11 bits are X's
- ✓ By this actual address range of EPROM can be determined

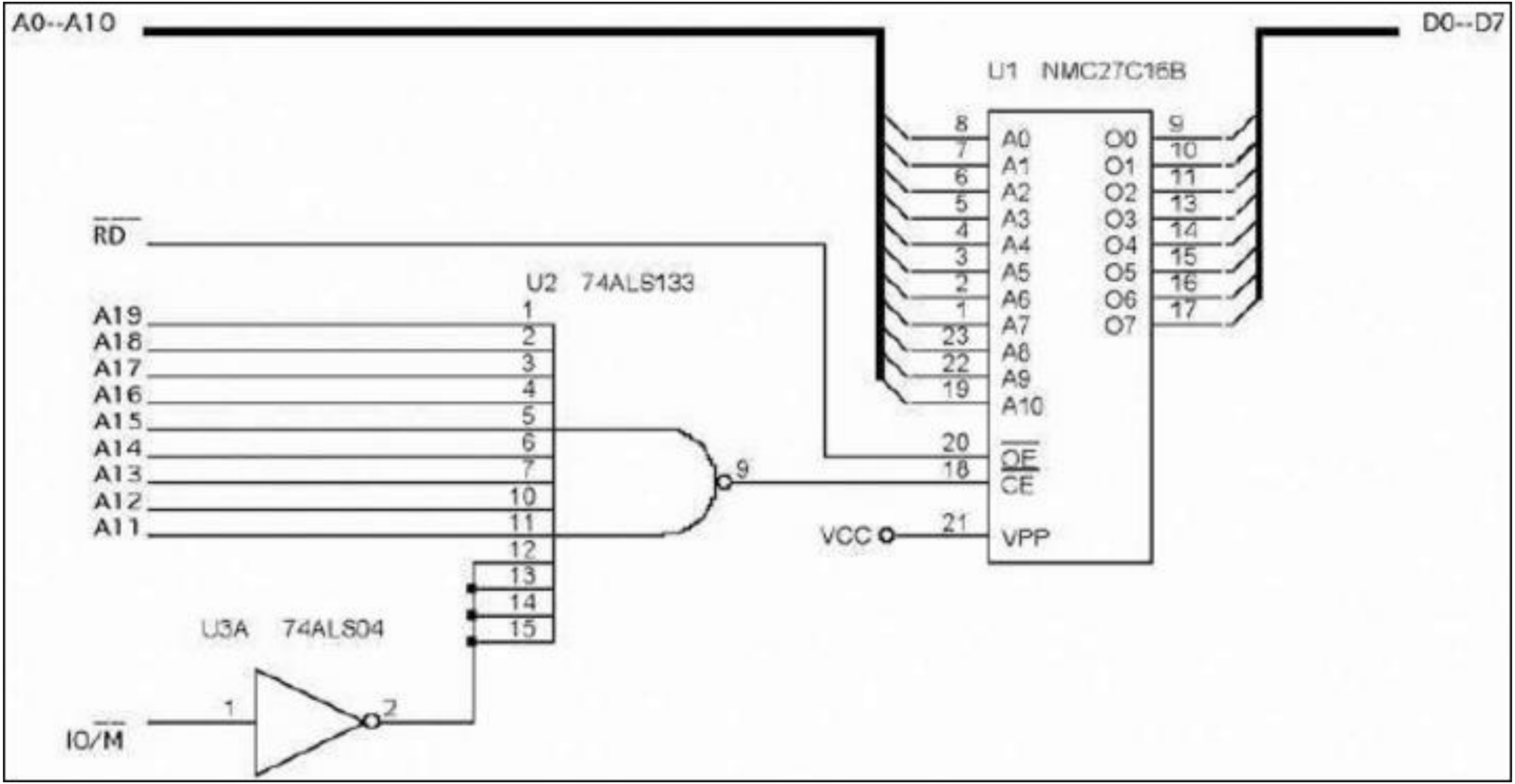
Ex: 1111 1111 1XXX XXXX XXXX

Starting Address 1111 1111 1000 0000 0000 = FF800H
Ending Address 1111 1111 1111 1111 1111 = FFFFFH

OR Ex: 0000 0000 0XXX XXXX XXXX

Starting Address 0000 0000 0000 0000 0000 = 00000H
Ending Address 0000 0000 0111 1111 1111 = 007FFH

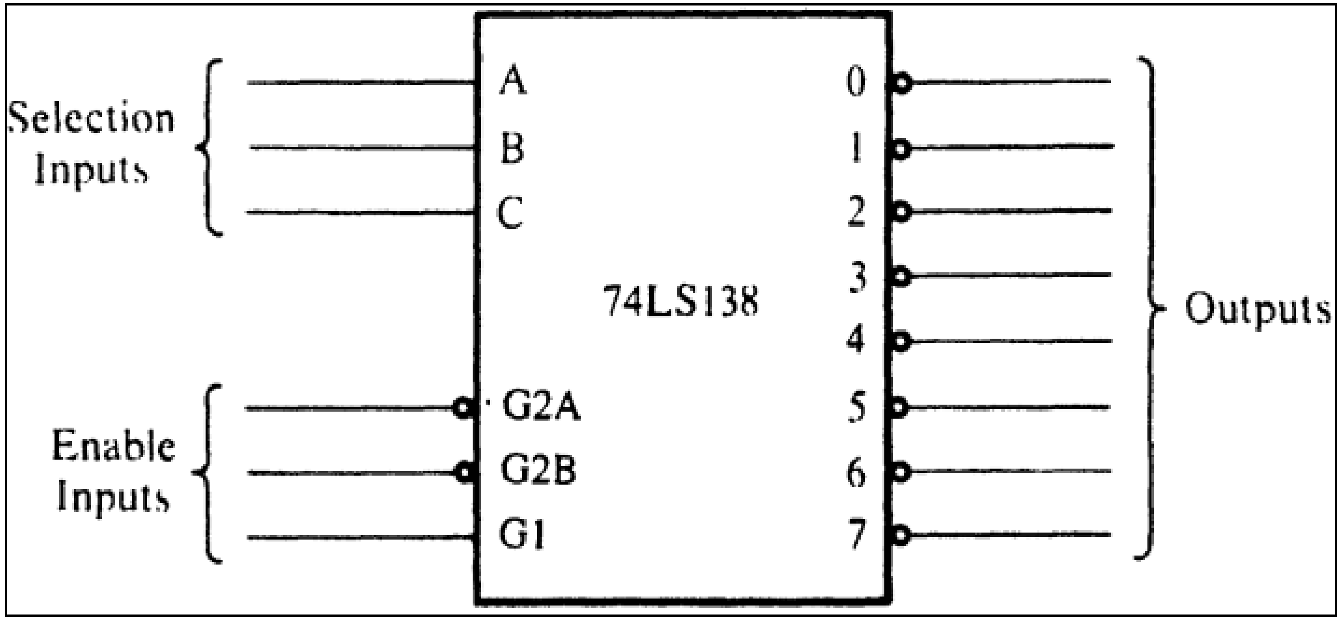
Simple NAND Gate Decoder:



Examples:

- Design an interface for 8088μp to connect a single 2716 (2kx8) EEPROM such that memory start at address FF800H.
- Design an interface for 8088μp to connect a three 27512 (64kx8) EEPROM such that memory start at address A0000H.

3-to-8 Line Decoder (74LS138):



Inputs						Outputs							
Enable			Select										
$\overline{G2A}$	$\overline{G2B}$	G1	C	B	A	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$
1	X	X	X	X	X	1	1	1	1	1	1	1	1
X	1	X	X	X	X	1	1	1	1	1	1	1	1
X	X	0	X	X	X	1	1	1	1	1	1	1	1
0	0	1	0	0	0	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	0

Imagine eight EPROM $\overline{CE}$ input pins connected to eight outputs of the decoder

2764 EPROM 8K x 8:

Design a 64kx8 section of memory for the 8088µp using 8kx8 (2764) EEPROMs. The starting address is F0000H. To enable decoder IC, A_{19}-A_{16} four connections must all be high.

	1111 XXXX XXXX XXXX XXXX	
Starting Address	1111 0000 0000 0000 0000 = F0000H	
Ending Address	1111 1111 1111 1111 1111 = FFFFFH	(its 64K-byte span of memory)

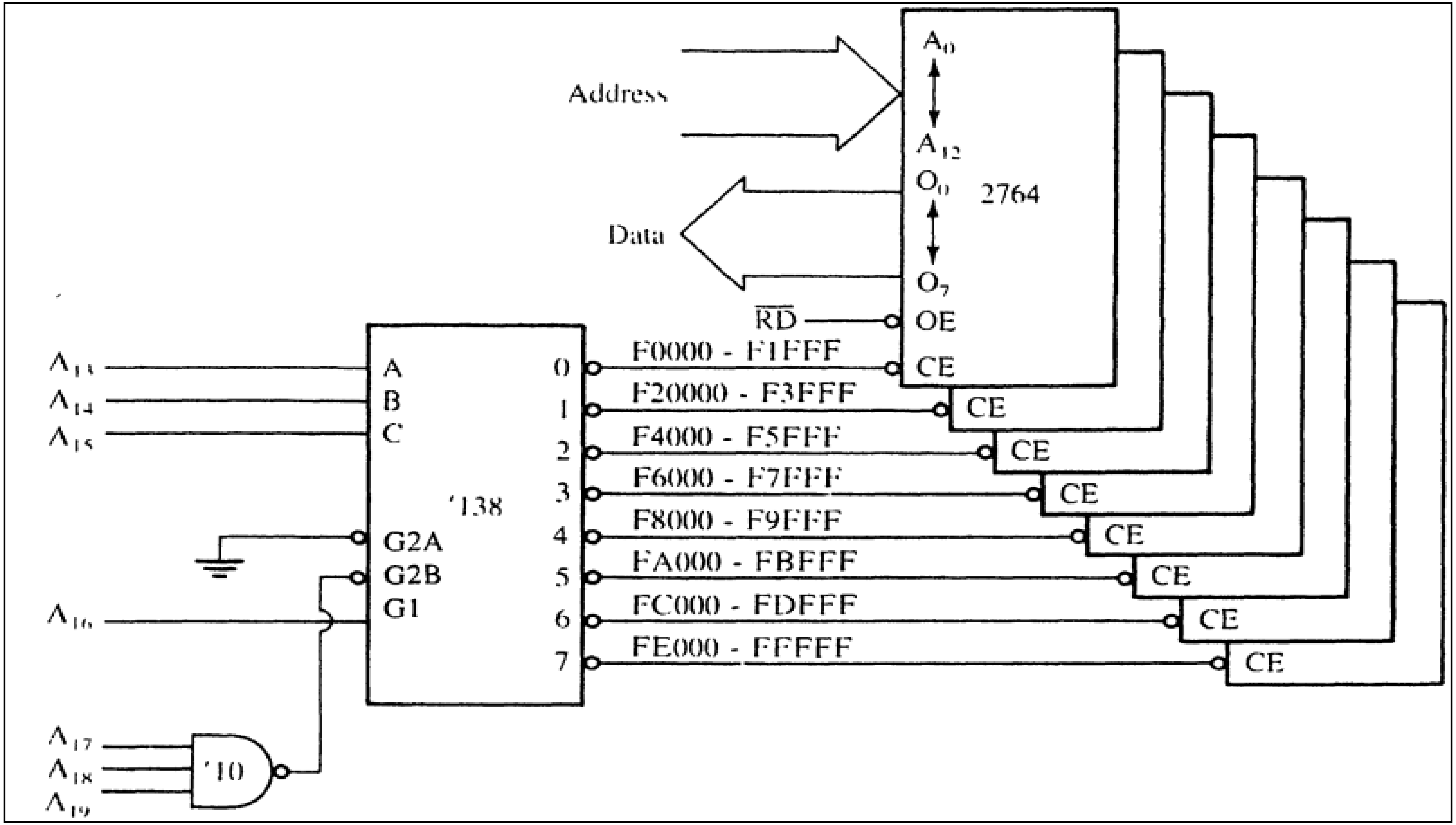
Each output range can be calculated as

	CBA
	1111 000X XXXX XXXX XXXX
	CBA
Starting Address	1111 0000 0000 0000 0000 = F0000H
Ending Address	1111 0001 1111 1111 1111 = F1FFFH
	CBA
Starting Address	1111 0010 0000 0000 0000 = F2000H
Ending Address	1111 0011 1111 1111 1111 = F3FFFH

2764 EPROM 8K x 8:

Design a 64kx8 section of memory for the 8088μp using 8kx8 (2764) EEPROMs.

The starting address is F0000H. To enable decoder IC, A_{19}-A_{16} four connections must all be high.



HOMEWORK

- Design a 64kx8 section of memory to a 8088µp using 8kx8 (2764) EEPROMs. The starting address is F0000H. Use Decoder 3-to-8
- Attach 4 64kx8 memory to a 8088µp EEPROMs. The starting address is 00000H. Use Decoder 3-to-8
- Attach 2 256kx8 memory to a 8088µp EEPROMs. The starting address is C0000H. Use Decoder 2-to-4

PLD (Programmable Logic Device) Programmer Decoder:

Arrays of logic elements that are programmable

Types of SPLD (Simple PLD)

- PLA (Programmable Logic Array)
- PAL (Programmable Array Logic)
- GAL (Gated Array Logic)

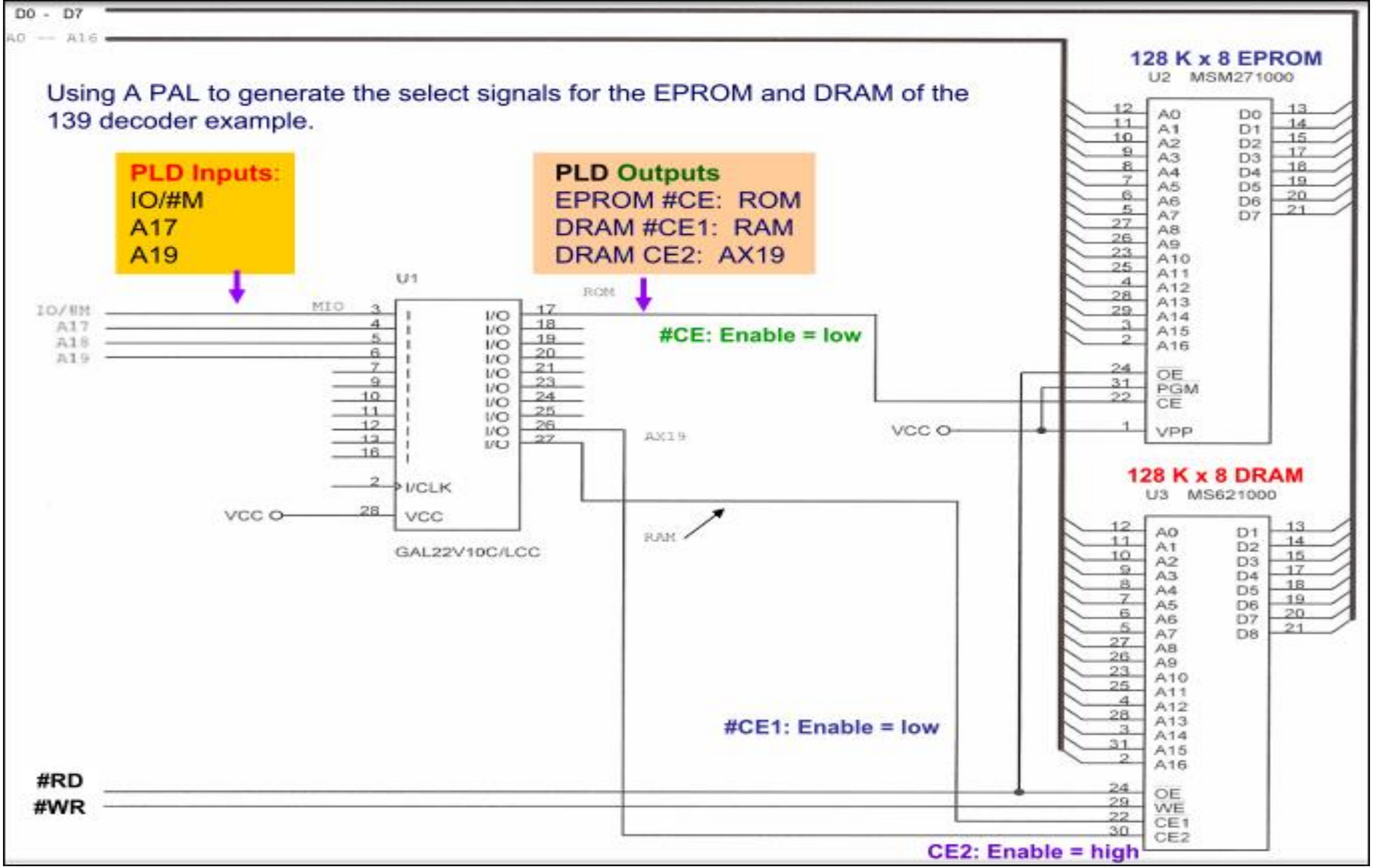
Other types ASIC (Application-Specific Integrated Circuits)

- CPLD (Complex Programmable Logic Array)
- FPGAs (Field Programmable Gate Arrays)
- FPICs (Field Programmable Interconnects)

A PAL is programed with HDL (Hardware Descriptive Language) or VHDL (Verilog HDL)

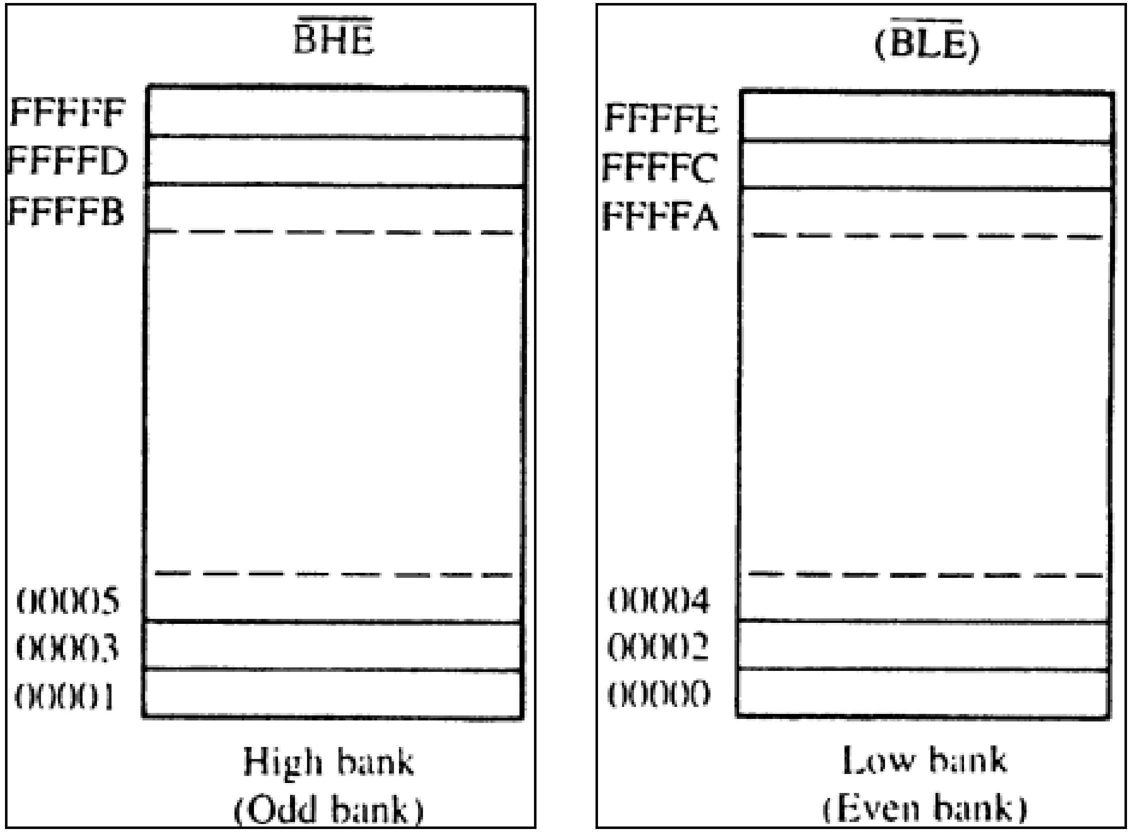
```vhdl
--VHDL code for the decoder of fig 10-17
library ieee;
use ieee.std_logic_1164.all;

entity DECODER_10_19 is

port (
		A19, A18, A17, MIO: in STD_LOGIC;
		ROM, RAM, AX19: out STD_LOGIC
	);
end;

architecture V1 of DECODER_10_19 is

begin
		ROM <= A19 or A18 or A17 or MIO;
		RAM <= not (A18 and A17 and (not MIO));
		AX19 <= not A19;
end V1;
```

Using A PAL to generate the select signals for the EPROM and DRAM of the 139 decoder example.



8086 Memory Interface:

- ✓ Address Bus is 20-bit wide
- ✓ Data bus is 16-bit wide
- ✓ $M/\overline{IO}$
- ✓ BHE (Bus High Enable)
- ✓ A_0 or BLE



8086 must be able to write data to any 16-bit location or any 8-bit location

16-bit bus must be divided into 2 separate sections (banks)

- ✓ Low Bank
- ✓ High Bank

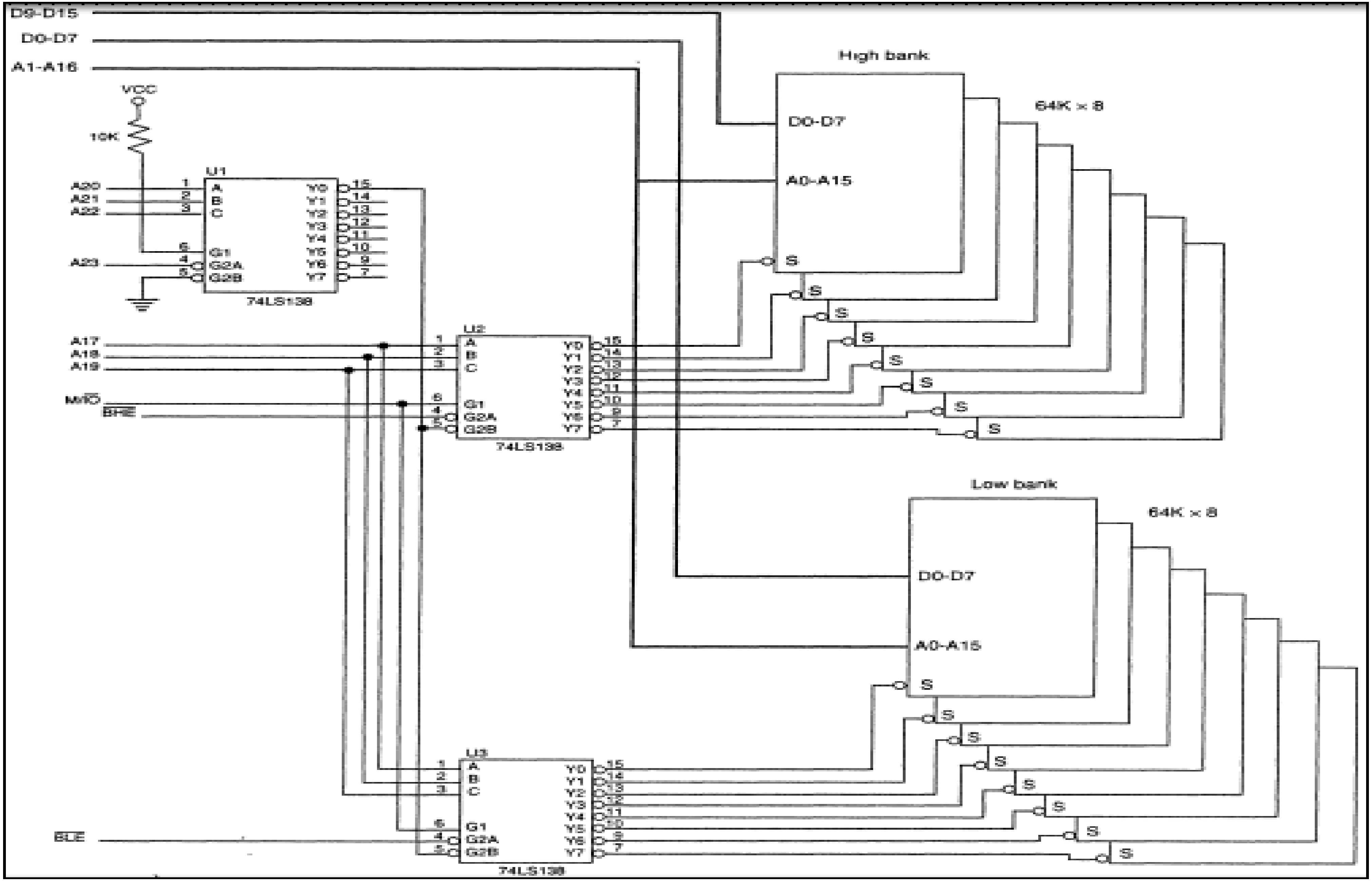
Bank Selection is accomplish in two ways:

- ✓ Separate Bank Decoders
- ✓ Separate Bank Write Strobes

$\overline{BHE}$	$\overline{BLE}$ (A_0)	Function
0	0	Both banks enabled for a 16-bit transfer
0	1	High bank enabled for an 8-bit transfer
1	0	Low bank enabled for an 8-bit transfer
1	1	No banks enabled

Separate Bank Decoders: (least effective way)

- two 74LS138 decoders used to select 64K RAM for 80386SX (24-bit address bus)
- Decoder U_2 has the BLE (A_0) attached to G2A
- Decoder U_3 has the BHE attached to G2A
- Decoder U_1 enables U_2 and U_3 for memory address range 000000H-0FFFFFH



D8-D15
D0-D7
A1-A16
High bank
64K × 8
D0-D7
A0-A15
S
VCC
10K
U1
A20
A21
A22
A23
A
B
C
G1
G2A
G2B
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
74LS138
U2
A17
A18
A19
M/IO
BHE
74LS138
Low bank
64K × 8
D0-D7
A0-A15
U3
BLE
74LS138

Separate Bank Write Strobes: (most effective way)

- Develop a separate write strobe for each memory bank
- This technique requires only 1-decoder